

Protechsoft Grooming notes Day3

1. What is JavaScript?

JavaScript (JS) is a scripting language used to make web pages interactive and dynamic.

Uses of JavaScript

- Form Validation
- Dynamic Content
- Event Handling
- API Calls
- DOM Manipulation

Ways to Write JavaScript

1. Internal JavaScript

```
<script>  
  console.log("Hello");  
</script>
```

2. External JavaScript

```
<script src="script.js"></script>
```

2. Variables

A variable is a container used to store data.

Ways to Declare Variables

```
var a = 10;  
let b = 20;  
const c = 30;
```

3. Difference Between var and let

var	let
Can be redeclared	Cannot be redeclared
Global scoped	Block scoped
Supports hoisting	Does not support hoisting

Example1:

```
var a = 10;
var a = 20;

let b = 10;
// let b = 20; Error
```

Example2:

```
{
  let x = 10;
  let y = 20;
  const z = 30;
  console.log(x); // 10
  console.log(y); // 20
  console.log(z); // 30
}
console.log(x); // 10
console.log(y); // Error
console.log(z); // Error
```

4. Variable Hoisting

Hoisting is JavaScript's behaviour of moving variable and function declarations to the top of their scope before execution.

Example

<pre>console.log(a); // undefined var a = 10;</pre>	Because JS treats it as: <pre>var a; console.log(a); a = 10;</pre>
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5. Data Types

Primitive Data Types

1. Number
2. String
3. Boolean
4. Undefined
5. Null
6. BigInt

Example

```
let age = 10;
let name = "Santanu";
let isStudent= true;

console.log(typeof age)           // number
console.log(typeof name)         // string
console.log(typeof isStudent)    // boolean
```

6. typeof Operator

Used to check the datatype of a variable.

Example

```
let age = 10;

console.log(typeof age); // number
```

7. Difference Between null and undefined

null	undefined
Assigned by the value null.	Variable declared but not assigned
typeof null is Object.	typeof null is undefined.

Example

```
let a = null;
let b;

console.log(a);           // null
console.log(typeof a);    // object
console.log(b);           // undefined
console.log(typeof b);    // undefined
```

8. String

A string is a sequence of characters enclosed with single quotes, double quotes or backticks.

Ways to Create String

Double Quotes

```
let name = "Santanu";
```

Single Quotes

```
let name = 'Santanu';
```

Backticks

```
let name = `Santanu`;
```

9. Template Literals / String Interpolation

Used to insert variables inside strings.

Syntax

`${variable}`

Example

```
let name = "Santanu";  
  
console.log(`Hello ${name}`); // Hello Santanu
```

Example 2

```
let a = 10  
let b = 20  
console.log(`the addition of ${a} and ${b} is ${a+b}`);  
  
// the addition of 10 and 20 is 30
```

10. String Methods

toUpperCase()

Converts string to uppercase and return one new string.

```
let s1 = "Hello";  
  
let s2 = s1.toUpperCase()  
console.log(s2); // HELLO
```

toLowerCase()

```
let s3 = "HeLLo"  
let s4 = s3.toLowerCase()  
console.log(s4) // hello
```

concat()

Used to combine strings and return new string.

```
let a = "Hello";  
let b = "World";  
  
console.log(a.concat( b)); //HelloWorld
```

trim()

Removes spaces from start and end.

```
let str = "  hello  ";  
  
console.log(str.trim()); // hello
```

slice()

Extracts part of a string.

It takes two parameters(start index and end index) it does not include the last index value.

```
let str = "JavaScript";  
  
console.log(str.slice(0,4)); //Java
```

11. Array

Array is a linear datastructure collection can store multiple values in a single variable.

Example

```
let arr = [10,20,30,40];  
  
console.log(arr[0]); // 10
```

12. Array Methods

push()

Adds element at the end.

```
let arr = [10,20];  
arr.push(30);  
console.log(arr); // [10,20,30]
```

pop()

Removes last element.

```
arr.pop();
```

unshift()

Adds element at beginning.

```
arr.unshift(5); // [5,10,20]
```

shift()

Removes first element.

```
arr.shift()
```

13. Function

A function is a block of code that performs a specific task.

Syntax

```
function greet(){  
  console.log("Hello");  
}
```

Calling Function

```
greet();
```

Output

Hello

14. Arrow Function

A shorter way to write functions.

Example

```
const greet = () =>{  
  console.log("Hello");  
}
```

```
greet();
```


15. Lexical Scoping

An inner function can access variables of its outer function.

Example

```
function outer(){  
    let a = 10;  
  
    function inner(){  
        console.log(a);  
    }  
  
    inner();  
}  
  
outer();  
// Output  
// 10
```

16. Closure

A closure is created when an inner function can access variables from its outer function even after the outer function has finished execution.

Example

```
function outer(){  
    let count = 0;  
  
    function inner(){  
        count++;  
        console.log(count);  
    }  
    return inner;  
}  
  
let result = outer();  
  
result(); // 1  
result(); // 2  
result(); // 3
```

17. For Loop

Used to execute a block of code repeatedly.

Syntax

```
for(initialization;condition;increment){  
  
}
```

Example

```
for(let i=1;i<=5;i++){  
  
    console.log(i);  
  
}
```

Output

```
1  
2  
3  
4  
5
```